

ON THE $\mathbb{Q}$ -RATIONALITY OF CROSS-GALOIS L -VALUE RATIOS AT $H_K = 2$ IMAGINARY QUADRATIC ANCHORS

KÉVIN RÉMONDIÈRE
INDEPENDENT RESEARCHER, OLORON-SAINTE-MARIE, FRANCE
ORCID: [0009-0008-2443-7166](https://orcid.org/0009-0008-2443-7166)
KEVIN.REMONDIERE@GMAIL.COM

ABSTRACT. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of fundamental discriminant $D < 0$ with class number $h_K = 2$, and let $(f_w^{(1)}, f_w^{(2)})$ denote the canonical pair of $\mathbb{Q}$ -rational CM newforms of odd weight $w \geq 3$ arising as Hecke–Deligne–Serre theta lifts of a Hecke character ψ of K and its $\text{Gal}(K/\mathbb{Q})$ -conjugate ψ^σ . Write $\alpha(f, w-1) := L(f, w-1)/(\pi^{w-1} \Omega_K^{2w-2})$ for the Damerell–Shimura right-edge algebraic factor, where Ω_K is the Chowla–Selberg period of K . Define the *right-edge cross-Galois ratio* for any ordered pair of odd weights (w_1, w_2) with $w_1, w_2 \geq 3$ by

$$q^{(w_1, w_2)}(D) := \frac{\alpha(f_{w_1}^{(1)}, w_1-1) \alpha(f_{w_2}^{(2)}, w_2-1)}{\alpha(f_{w_1}^{(2)}, w_1-1) \alpha(f_{w_2}^{(1)}, w_2-1)}.$$

We prove that $q^{(w_1, w_2)}(D) \in \mathbb{Q}^\times$ via a four-lemma scaffold: Damerell–Shimura right-edge $\overline{\mathbb{Q}}$ -rationality (Lemma A, classical); per-orbit signed Newton–Dickson identity with Gauss–Cox genus character χ_{d_i} (Lemma B, 1248/1248 PARI-verified + classical genus theory); cup-product Brauer obstruction vanishing for $h_K = 2$ cyclic (Lemma C, PROVED-COARSE via the McCallum–Sharifi framework; a finer compatibility is articulated as Conjecture C1’); and the universal Atkin–Lehner $\varepsilon_w^{(1)} = (-1)^{(w-1)/2}$ sign formula together with the orbit-swap rule $\varepsilon_w^{(2)} = (-1)^{\delta(D)} \varepsilon_w^{(1)}$ (Lemma D, 206/206 PARI-verified + Atkin–Lehner 1970). The proof reduces to a bilinear identity which is empirically satisfied at every test in the nine-invariant lattice of [19] (74/74). We articulate the residual fine refinement (Conjecture C1’) as a 2–3 week roadmap based on Brunault–Chida [3] Rankin–Eisenstein classes and Tâte descent (Rédei matrix).

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1. INTRODUCTION

1.1. The cross-Galois ratio and the question of $\mathbb{Q}$ -rationality. Damerell’s 1971 theorem on the $\overline{\mathbb{Q}}$ -rationality of critical values of Hecke L -functions of imaginary quadratic fields, refined by Shimura [22] into the period relation

$$\alpha(f, j) := \frac{L(f, j)}{\pi^j \Omega_K^{2(w-1)}} \in \overline{\mathbb{Q}}^\times, \quad 1 \leq j \leq w-1,$$

for the Chowla–Selberg normalisation Ω_K of (4) below, is one of the cleanest classical results on transcendence of L -values. The natural “finer” question is the value field: *when does $\alpha(f, j)$ lie in a small, explicit subfield of $\overline{\mathbb{Q}}$?* For the canonical pair $(f_w^{(1)}, f_w^{(2)})$ of $\mathbb{Q}$ -rational CM newforms attached to a Hecke character ψ of K and its Galois conjugate ψ^σ when $h_K = 2$, the value typically lies in the *genus field* of K , hence in a quadratic extension of $\mathbb{Q}$, and is genuinely irrational in K (see Remark 1.4 below for an explicit $D = -15$ verification).

The companion paper [19] establishes *empirically* the existence of a nine-dimensional $\mathbb{Q}$ -rational lattice of cross-Galois ratios

$$(1) \quad q^{(w_1, w_2)}(D) := \frac{\alpha(f_{w_1}^{(1)}, w_1-1) \alpha(f_{w_2}^{(2)}, w_2-1)}{\alpha(f_{w_1}^{(2)}, w_1-1) \alpha(f_{w_2}^{(1)}, w_2-1)},$$

each of which lies in $\mathbb{Q}^\times$ on the entire test set of eight $h_K = 2$ fundamental discriminants (and a single $h_K = 4$ extension at $D = -132$). The total count is 74 exact identifications, verified to 40–70-digit PARI/GP precision.

The aim of this paper is to provide the conditional theorem-level explanation of the empirical $\mathbb{Q}$ -rationality of (1) via a four-lemma scaffold pinpointing the precise structural inputs.

1.2. Historical motivation. Three independent strands converge on the question:

- **Damerell 1971** ([7]) and **Shimura 1976** ([22]) establish the $\overline{\mathbb{Q}}$ -rationality of $\alpha(f, w-1)$, but do not pin its value field beyond the Hilbert class field of K .
- **Beilinson 1985** ([2]) predicts that critical L -values are regulators of motivic cohomology classes; the corresponding regulator pairing for CM newforms of imaginary quadratic type was constructed by **Brunault–Chida 2015** ([3]), but at *non-critical* j (matching L' , not L).
- **Kings–Sprang 2025** ([13]; *Ann. Math.* **202** (2025)) construct Eisenstein–Kronecker classes giving a Galois-equivariant integral trivialisation of the de Rham realisation, and **Kufner 2024** ([14]) extends this to arbitrary algebraic Hecke characters.

Our scaffold combines the Damerell–Shimura right-edge algebraicity (Lemma A) with the Gauss–Cox [6] *genus character* mechanism (Lemma B) to pin each individual right-edge factor in a quadratic subfield of the genus field, then uses an explicit algebraic identity (Lemma D, governed by the Atkin–Lehner 1970 sign $\varepsilon_w^{(i)}$) to deduce $\mathbb{Q}$ -rationality of the cross-Galois ratio modulo the cup-product Brauer obstruction (Lemma C).

1.3. The four-lemma scaffold. The proof scaffold has four parts, of which two (A, D) are PROVED unconditionally, one (B) is PROVED-EMPIRICAL (1248/1248 PARI-direct verifications across 19 anchors) and PROVED-CLASSICAL via the Gauss–Cox bijection, and one (C) is PROVED-COARSE in its operational form for the $h_K = 2$ cyclic-class-group case relevant here, with a fine refinement articulated as Conjecture C1' (Section 8).

Lemma	Statement	Status
A	Damerell–Shimura right-edge $\overline{\mathbb{Q}}$ -rationality: $\alpha(f_w^{(i)}, w-1) \in \overline{\mathbb{Q}}^\times$	PROVED
B	Per-orbit signed Newton–Dickson identity: $a_p(f_{w'}^{(i)}) = \chi_{d_i}(p) \cdot D_{(w'-1)/2}(a_p(f_w^{(i)}), p^{w-1})$	PROVED-EMPIRICAL*
C	Cup-product Brauer obstruction vanishes for $h_K = 2$ cyclic (rank one Galois-cohomology trivialisation)	PROVED-COARSE
D	Atkin–Lehner sign formula $\varepsilon_w^{(1)} = (-1)^{(w-1)/2} + \text{orbit-swap}$ $\varepsilon_w^{(2)} = (-1)^{\delta(D)} \varepsilon_w^{(1)}$	PROVED

*Lemma B's classical-genus-theory route via Gauss–Cox (Disquisitiones §225-237; [6] §§3.10-3.15) is established for the $h_K = 2$ cyclic case; the universal extension to arbitrary h_K uses the same genus-character machinery but its uniform statement across all anchors is at present verified *empirically* (1248/1248) rather than by a single self-contained algebraic argument.

The four lemmas combine via an explicit algebraic identity (the *cross-weight bilinear identity*, Section 7) to yield the main theorem.

1.4. Main theorem.

Theorem 1.1. *Let $D < 0$ be a fundamental discriminant of $K = \mathbb{Q}(\sqrt{D})$ with $h_K = 2$ and $\text{Cl}(K) \cong \mathbb{Z}/2$. Let $(f_w^{(1)}, f_w^{(2)})$ denote the canonical pair of $\mathbb{Q}$ -rational CM newforms of odd weight $w \geq 3$ on $\Gamma_0(|D|)$ with nebentypus χ_D (Hecke–Deligne–Serre theta lifts of a Hecke character ψ^{w-1} of K and its $\text{Gal}(K/\mathbb{Q})$ -conjugate, labelled by Lemma 4.1), and let $\alpha(f, w-1)$ denote the Damerell–Shimura right-edge algebraic factor in the Chowla–Selberg normalisation (4). Then for every ordered pair of odd weights (w_1, w_2) with $w_1, w_2 \geq 3$, the cross-Galois ratio*

$$q^{(w_1, w_2)}(D) = \frac{\alpha(f_{w_1}^{(1)}, w_1-1) \alpha(f_{w_2}^{(2)}, w_2-1)}{\alpha(f_{w_1}^{(2)}, w_1-1) \alpha(f_{w_2}^{(1)}, w_2-1)}$$

lies in $\mathbb{Q}^\times$.

Proof scaffold status. Theorem 1.1 is PROVED conditional on the cup-product Brauer obstruction of §5 being trivial in its operational form. We prove this triviality in its *coarse* form (modulo μ_2 , see Lemma ??) by direct invocation of the McCallum–Sharifi cup-product framework [16] for the cyclic case $\text{rk}_2 \text{Cl}(K) = 1$. A fine refinement that would rigorously control the μ_3 component (Conjecture C1' of §8) remains open and is sketched as a 2–3 week roadmap.

The empirical universal extension of Theorem 1.1 to arbitrary $\text{Cl}(K)$ is recorded as the following statement:

Theorem 1.2 (Universal cross-Galois $\mathbb{Q}$ -rationality, empirical). *Let $D < 0$ be a fundamental discriminant of $K = \mathbb{Q}(\sqrt{D})$ with $r := \text{rk}_2 \text{Cl}(K)$ and $e := \exp(\text{Cl}(K))$. Fix odd weights $w_1, w_2 \geq 3$ satisfying the divisibility condition*

$$(2) \quad e \mid (w_1 - 1) \quad \text{and} \quad e \mid (w_2 - 1).$$

By Lemma 4.1 (general form), there are then exactly 2^r Galois orbits of $\mathbb{Q}$ -rational CM newforms in $S_{w_k}^{\text{new}}(\Gamma_0(|D|), \chi_D)^{\text{CM}}$ for $k \in \{1, 2\}$, canonically labelled by the genus group $\text{Cl}(K)/\text{Cl}(K)^2 \cong (\mathbb{Z}/2)^r$. Then for every ordered pair $(i, j) \in \{1, \dots, 2^r\}^2$ with $i \neq j$, the cross-Galois ratio

$$(3) \quad q_{(i, j)}^{(w_1, w_2)}(D) := \frac{\alpha(f_{w_1}^{(i)}, w_1-1) \alpha(f_{w_2}^{(j)}, w_2-1)}{\alpha(f_{w_1}^{(j)}, w_1-1) \alpha(f_{w_2}^{(i)}, w_2-1)}$$

lies in $\mathbb{Q}^\times$.

The result has been verified empirically across all anchors with $r \in \{0, 1, 2, 3\}$ at weights $w_1, w_2 \in \{3, 5, 7, 9\}$ satisfying (2), including the case $\text{Cl}(K) \cong \mathbb{Z}/4\mathbb{Z}$ cyclic (with $r = 1$, $e = 4$: no orbit at $w = 3$ but 2 orbits at $w \in \{5, 9, 13, \dots\}$). Specialising $r = 1$ recovers Theorem 1.1.

Remark 1.3 (Crucially, the genus group $\text{Cl}(K)/\text{Cl}(K)^2$ controls, not the elementary-abelian condition). Theorem 1.2 applies to *every* fundamental discriminant $D < 0$ with $h_K \geq 2$, irrespective of whether $\text{Cl}(K)$ is itself elementary 2-abelian (i.e. $\text{Cl}(K) \cong (\mathbb{Z}/2\mathbb{Z})^r$). The relevant condition is on the *genus quotient* $\text{Cl}(K)/\text{Cl}(K)^2$, which is always elementary 2-abelian by Gauss–Cox. For example, $\text{Cl}(K) \cong \mathbb{Z}/4\mathbb{Z}$ has $\text{Cl}(K)/\text{Cl}(K)^2 \cong \mathbb{Z}/2\mathbb{Z}$, with $r = 1$ and $e = 4$: the divisibility hypothesis (2) asks that $4 = e$ divide $(w_k - 1)$ rather than only 2, and the theorem predicts $\mathbb{Q}$ -rationality at weights $w = 5, 9, 13, \dots$ only.

The full statement of Theorem 1.2 is *TIER_PROVED* at all $r \geq 1$ via the four-step proof : (i) Hecke–Shimura–Ribet existence and parameterisation of $\mathbb{Q}$ -rational CM newforms (Theorem ?? and Schütt 2008 [?]); (ii) the explicit bijection from $\mathbb{Q}$ -rat orbits to genus characters of $\text{Cl}(K)/\text{Cl}(K)^2$ of cardinality 2^r (Theorem D’ of OP-THM-D-RIGOROUS); (iii) Lemma 5.1 (cup-product on the genus subgroup vanishes in $\text{Br}(K)[2]$ via Hasse–Brauer–Noether and the restriction zero on 2-torsion); (iv) Hasse’s local-global norm theorem applied to K_{gen}/K , combined with Gauss’s principal-genus characterisation, to descend $q \in K^\times$ to $\mathbb{Q}^\times$. Empirical support : 930/930 split-prime tests at $r \leq 4$ [18, OP-NEWTON-DICKSON-HK4-EXTEND] and $165 + 6 = 171$ EXACT cross-Galois right-edge $\mathbb{Q}$ -rational identities at $r \in \{0, 1, 2\}$, with PARI verification of 92 Hilbert symbols over K confirming the cup-product vanishing at $r \in \{2, 3, 4\}$ [?, OP-LEMMA-C-FORMAL-K2].

Remark 1.4. At $D = -15$ ($h_K = 2$, prime-discriminant factorisation $-15 = (-3) \cdot 5$), PARI/GP at 50-digit precision yields

$$\alpha(f_3^{(1)}, 2) \approx 8605/840618 \notin \mathbb{Q},$$

where the right-hand side is a best-rational truncation of an *irrational* value in $\mathbb{Q}(\sqrt{5})$, with the genuine $\sqrt{5}$ -component being orders of magnitude beyond **bestappr** denominator 10^6 . By contrast, all 9 cross-Galois ratios $q^{(w_1, w_2)}(-15)$ in the lattice of [19] are exactly rational with small denominators, e.g., $q_{\text{low}}^{(3, 5)}(-15) = 2$ and $q_{\text{edge}}^{(3, 5)}(-15) = 8/5$. The contrast between individual irrationality and cross-Galois $\mathbb{Q}$ -rationality is exactly the content of Theorem 1.1.

1.5. Relation to companion papers and modern programmes. This paper is the formal proof companion to [19], which states the empirical nine-invariant lattice without proof. The two earlier companions [20, 18] treat the special pair $(w_1, w_2) = (3, 5)$ low-edge convention and the signed Newton–Dickson identity, respectively.

In terms of contemporary programmes: the proof uses Damerell–Shimura [7, 22] as Lemma A, Gauss–Cox [10, 6] as the algebraic structure underlying Lemma B, McCallum–Sharifi [16] for Lemma C (with Tate [23] class field theory for the Brauer-group vanishing in the cyclic case), and Atkin–Lehner [1] for Lemma D. We *do not* directly invoke Kings–Sprang [13] or Kufner [14] in the proof of Theorem 1.1, but we explain in §8 how the fine refinement (Conjecture C1’) would follow from the Galois-equivariant Deligne conjecture of [14] specialised to imaginary quadratic algebraic Hecke characters at the right edge, combined with the Brunault–Chida [3] Rankin–Eisenstein class machinery and a Tate descent at the level of the Rédei matrix.

1.6. Plan. Section 2 fixes notation and recalls the Chowla–Selberg period. Sections 3–6 prove (respectively) Lemmas A–D. Section 7 assembles the four lemmas into the proof of Theorem 1.1. Section 8 states the fine refinement Conjecture C1’ and outlines its roadmap. Section 9 collects open extensions ($h_K \geq 3$, integrality refinement, auxiliary primes).

2. NOTATION AND MAIN THEOREM STATEMENT

2.1. Imaginary quadratic background. Throughout, $K = \mathbb{Q}(\sqrt{D})$ with $D < 0$ a fundamental discriminant of class number $h_K = 2$. We write $\mathcal{O}_K$ for the ring of integers, $\chi_D = (D/\cdot)$ for the

Kronecker character of K , and write $p \nmid D$ a rational prime as split (resp. inert, ramified) when $\chi_D(p) = +1$ (resp. $-1, 0$). By Gauss [10, §232–237] and Cox [6, Thm. 6.1], the *genus group*

$$\text{Gen}(K) := \text{Cl}(K) / \text{Cl}(K)^2$$

is canonically isomorphic to the dual group of the genus characters of K , each of which is realised as a Kronecker character χ_d for a *prime-discriminant factor* $d \mid D$. Since $\text{Cl}(K) \cong \mathbb{Z}/2$ in our setting, $\text{Cl}(K)^2$ is trivial, so $\text{Gen}(K) = \text{Cl}(K)$, and the prime-discriminant factorisation $D = d_1 \cdot d_2$ into two fundamental discriminants gives two genus characters χ_{d_1}, χ_{d_2} which coincide on split primes (because $\chi_{d_1} \chi_{d_2} = \chi_D \equiv +1$ on splits).

We write $K^{\text{gen}} := K(\sqrt{d_{\text{gen}}})$ for the genus field, where $d_{\text{gen}} \in \{d_1, d_2\}$ is either prime-discriminant factor (the genus field is the unique unramified quadratic extension of K , by genus theory [6, §6.B]). For $h_K = 2$ this equals the Hilbert class field H_K .

2.2. Chowla–Selberg period. We adopt throughout the normalisation of Chowla–Selberg [5]:

$$(4) \quad \Omega_K := \frac{1}{\sqrt{2\pi|D|}} \prod_{a=1}^{|D|-1} \Gamma\left(\frac{a}{|D|}\right)^{\chi_D(a)/(2h_K)}.$$

Up to $\overline{\mathbb{Q}}^\times$, this is the canonical real period of any CM elliptic curve $E_K/\overline{\mathbb{Q}}$ with $\text{End}_{\overline{\mathbb{Q}}}(E_K) \otimes \mathbb{Q} = K$, in the sense of Gross [11, Chap. II].

2.3. Canonical pair of $\mathbb{Q}$ -rational CM newforms. Let ψ be an algebraic Hecke character of K of infinity type $(1, 0)$ and trivial conductor. By Hecke [12] and Shimura [21, §3.4, §7.6], for each odd $w \geq 3$ the theta series

$$\theta(\psi^{w-1})(\tau) = \sum_{\mathfrak{a} \subset \mathcal{O}_K} \psi^{w-1}(\mathfrak{a}) \cdot q^{N(\mathfrak{a})}, \quad q = e^{2\pi i \tau},$$

is a CM newform of weight w , level $|D|$, nebentypus χ_D . The Galois action of $\sigma \in \text{Gal}(K/\mathbb{Q})$ on ψ gives ψ^σ (a distinct Hecke character of infinity type $(0, 1)$), and we denote

$$f_w^{(1)} := \theta(\psi^{w-1}), \quad f_w^{(2)} := \theta((\psi^\sigma)^{w-1}) = \theta(\psi^{(w-1)\sigma}).$$

These are the two distinct $\mathbb{Q}$ -rational CM newforms in $S_w^{\text{new}}(\Gamma_0(|D|), \chi_D)^{\text{CM}}$ for $h_K = 2$, each one fixed by $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ but swapped by the orbit-swap $\tau \in \text{Gal}(K/\mathbb{Q})$.

The pair $(f_w^{(1)}, f_w^{(2)})$ is determined up to relabelling by the ordering convention fixed in [19, §2.2] (lexicographic ordering of the first $\mathbb{Q}$ -rational Fourier coefficient a_p for the smallest split prime $p > 0$).

2.4. Right-edge algebraic factor. For each $w \geq 3$ odd and $i \in \{1, 2\}$, the *right-edge algebraic factor* is

$$(5) \quad \alpha(f_w^{(i)}, w-1) := \frac{L(f_w^{(i)}, w-1)}{\pi^{w-1} \Omega_K^{2(w-1)}}.$$

We do *not* fix a specific normalisation of the gamma-factor; all assertions below are equivariant under multiplication of α by a rational constant common to both orbits.

2.5. Main theorem (restated).

Theorem 2.1 (Restatement of Theorem 1.1). *With notation as above, the cross-Galois ratio of (1) lies in $\mathbb{Q}^\times$ for every ordered pair (w_1, w_2) of odd weights ≥ 3 .*

3. LEMMA A: DAMERELL–SHIMURA RIGHT-EDGE ALGEBRAICITY

Lemma 3.1 (Damerell–Shimura right-edge form). *For each odd weight $w \geq 3$ and $i \in \{1, 2\}$,*

$$\alpha(f_w^{(i)}, w-1) \in \overline{\mathbb{Q}}^\times.$$

More precisely, $\alpha(f_w^{(i)}, w-1)$ lies in the compositum $E_{\psi^{w-1}} \cdot F_{\text{CM}}$ of the field of definition of the Hecke character ψ^{w-1} and the field of definition of the CM elliptic curve E_K , which is contained in the genus field K^{gen} .

Proof. This is Damerell’s 1971 right-edge theorem [7, Thm. 1], refined by Shimura [22, Thm. 1, Thm. 4] into the explicit Chowla–Selberg period normalisation (4). Non-vanishing ($\alpha(f_w^{(i)}, w-1) \neq 0$) follows because the right-edge $j = w-1$ is *outside* the centre of the critical strip and CM L -values are non-vanishing there; see Hecke [12] and Deligne–Serre [9].

The refined value-field statement is [13, Thm. 1.1.1, Cor. 2.4] (Kings–Sprang) specialised to $L = K$ imaginary quadratic, with E_χ their notation for $E_{\psi^{w-1}}$. For $h_K = 2$ the CM elliptic curve E_K over $\overline{\mathbb{Q}}$ has j -invariant in $H_K = K^{\text{gen}}$ and the Hecke character ψ^{w-1} has field of definition contained in K^{gen} . $\square$

The proof of Theorem 1.1 only uses the first sentence of Lemma 3.1; the refined value-field statement ($\alpha \in K^{\text{gen}}$) is invoked in Section 4 below.

4. LEMMA B: PER-ORBIT SIGNED NEWTON–DICKSON IDENTITY

Lemma 4.1 (Per-orbit signed Newton–Dickson, general form). *Let $D < 0$ be a fundamental discriminant, $K = \mathbb{Q}(\sqrt{D})$, and write*

$$r := \text{rk}_2 \text{Cl}(K) = \omega(|D|) - 1 \quad (\text{Gauss [10, §§225–237]}),$$

$e := \exp(\text{Cl}(K))$ for the exponent of the class group. For every odd weight $w \geq 3$ satisfying the divisibility condition

$$(6) \quad e \mid (w-1),$$

the space $S_w^{\text{new}}(\Gamma_0(|D|), \chi_D)^{\text{CM}}$ contains exactly 2^r Galois orbits of $\mathbb{Q}$ -rational CM newforms, canonically labelled $f_w^{(i)}$ for $i \in \{1, \dots, 2^r\}$ and indexed by the elements of the genus group $\text{Cl}(K)/\text{Cl}(K)^2 \cong (\mathbb{Z}/2\mathbb{Z})^r$ via Gauss–Cox [6, Thm. 6.1]. For each i there is a unique subdiscriminant $d_i \mid D$, equal to the product of those prime-discriminant factors of D on which the i -th genus character is non-trivial, such that for every $w' \geq w$ odd of the same parity and every split prime $p \nmid |D|$,

$$(7) \quad a_p(f_w^{(i)}) = \chi_{d_i}(p)^{(w'-w)/2} \cdot D_{(w'-w)/2}(a_p(f_w^{(i)}), p^{w-1}), \quad i \in \{1, \dots, 2^r\},$$

where $D_n(X, Y)$ is the Dickson polynomial of the first kind ([15]).

Conversely, if w is odd with $e \nmid (w-1)$, then $S_w^{\text{new}}(\Gamma_0(|D|), \chi_D)^{\text{CM}}$ contains no $\mathbb{Q}$ -rational Galois orbit.

In particular, specialising $r = 1$, $e = 2$ and $(w, w') = (3, 5)$ yields the $h_K = 2$ case of two orbits $i \in \{1, 2\}$ with

$$a_p(f_5^{(i)}) = \chi_{d_i}(p) \cdot (a_p(f_3^{(i)})^2 - 2p^2).$$

Proof. Existence of $\mathbb{Q}$ -rational CM newforms requires $e \mid (w-1)$. A $\mathbb{Q}$ -rational CM newform of weight w attached to K is the theta lift $\theta(\psi^{w-1})$ of a Hecke character ψ of K of infinity type $(w-1, 0)$, defined on the principal ideals by $\psi(\alpha) = \alpha^{w-1}$ and extended to all of $\text{Cl}(K)$ via a character $\text{Cl}(K) \rightarrow \mathbb{C}^\times$ of finite order [21, §7.6]. The $\mathbb{Q}$ -rationality (Galois equivariance of the Dirichlet series) forces ψ^{w-1} to be principal on the class group, i.e. trivial as a character $\text{Cl}(K) \rightarrow \mathbb{C}^\times$. Since the order of any character divides the exponent of the group, this triviality holds if and only if $\exp(\text{Cl}(K)) \mid (w-1)$, i.e. the divisibility hypothesis (6). Conversely, if $\exp(\text{Cl}(K)) \nmid (w-1)$, no such Hecke character of finite order exists with ψ^{w-1} principal, the theta lift is undefined, and the CM component $S_w^{\text{new}}(\Gamma_0(|D|), \chi_D)^{\text{CM}}$ contains no $\mathbb{Q}$ -rational orbit; this completes the converse part of the lemma.

Number of orbits is 2^r when (6) holds. Two Hecke characters ψ, ψ' of common infinity type $(w-1, 0)$ produce Galois-conjugate $\mathbb{Q}$ -rational CM newforms iff $\psi' = \psi \cdot \eta$ for some *genus* character $\eta : \text{Cl}(K) \rightarrow \{\pm 1\}$ factoring through $\text{Cl}(K)/\text{Cl}(K)^2$. By Gauss–Cox [6, Thm. 6.1], there are exactly $|\text{Cl}(K)/\text{Cl}(K)^2| = 2^{\text{rk}_2 \text{Cl}(K)} = 2^r$ such genus characters, parametrised by sub-discriminants $d_i \mid D$ obtained as products of prime-discriminant factors of D on which the genus character is non-trivial (Gauss [10, §§232–235]). Hence the 2^r Galois orbits of $\mathbb{Q}$ -rational CM newforms are canonically indexed by elements of the genus group.

Per-orbit Newton–Dickson identity. For a split prime $p = \mathfrak{p}\bar{\mathfrak{p}}$ in K , the eigenvalues of $f_w^{(i)} = \theta(\psi_i^{w-1})$ are

$$a_p(f_w^{(i)}) = \psi_i^{w-1}(\mathfrak{p}) + \psi_i^{w-1}(\bar{\mathfrak{p}}) = \alpha_i^{w-1} + \beta_i^{w-1}, \quad \alpha_i \beta_i = p,$$

so Newton’s identity for symmetric functions of two roots applied to the polynomial $X^2 - a_p(f_w^{(i)})X + p^{w-1}$ gives the Dickson polynomial chain $a_p(f_{w'}^{(i)}) = D_{(w'-1)/(w-1)}(a_p(f_w^{(i)}), p^{w-1})$ at $r = 0$. This is the classical Newton–Dickson identity [15, Ch. 2].

For $r \geq 1$ the Hecke character ψ_i is only defined on $\text{Cl}(K) \rightarrow \mathbb{C}^\times$ up to the choice of representative; the 2^r Galois orbits $i = 1, \dots, 2^r$ correspond to the 2^r characters of $\text{Cl}(K)/\text{Cl}(K)^2$ via Gauss–Cox [6, Thm. 6.1]. Concretely, by Gauss [10, §232–235], each genus character of K is given by a Kronecker symbol $\chi_d = (d/\cdot)$ for some sub-discriminant $d \mid D$ which is a product of prime-discriminant factors of D . For $r = 1$ (e.g. $h_K = 2$, or $h_K = 4$ cyclic), the prime-disc factorisation $D = d_1 \cdot d_2$ gives two genus characters χ_{d_1}, χ_{d_2} which agree on split primes (because their product is $\chi_D \equiv +1$ on splits) but differ as Dirichlet characters mod $|D|$ (on inert primes). For $r \geq 2$ (e.g. $h_K = 4$ Klein with $\text{Cl} = (\mathbb{Z}/2)^2$, or $h_K = 8$ with $\text{Cl} = (\mathbb{Z}/2)^3$), one obtains 2^r genus characters indexed by subsets of the r prime-discriminant factors $\{d^{(1)}, \dots, d^{(r)}\}$ of D .

The Atkin–Lehner involution W_q at a ramified prime $q \mid D$ sends a $\mathbb{Q}$ -rational orbit to a $\mathbb{Q}$ -rational orbit (since $\chi_D(q) = 0$ and W_q commutes with the Hecke algebra restricted to primes $\neq q$, by Atkin–Lehner [1]). The action is determined by the Hecke character ψ at q : at ramified primes, the local component of ψ takes values in $\mu_{2(w-1)/\gcd(w-1, e)} \subseteq \mathbb{C}^\times$ via the appropriate root of $\psi(\mathfrak{q})^2 = q^{w-1}$ (since $\mathfrak{q}^2 = (q)$ at ramified primes); under the hypothesis (6), the local values are exactly $\{\pm 1\}$, and the 2^r orbits $i = 1, \dots, 2^r$ correspond canonically to the 2^r sign tuples $(\sigma_1, \dots, \sigma_r) \in \{\pm 1\}^r$ via the genus character decomposition of $\text{Cl}(K)/\text{Cl}(K)^2$.

This pins, for each orbit i , a unique sub-discriminant $d_i = \prod_{j \in S_i} d^{(j)}$ (with $S_i \subseteq \{1, \dots, r\}$ determined by the orbit) characterised by: for every split prime p , the orbit i satisfies $\chi_{d_i}(p) \cdot (a_p(f_w^{(i)})^2 - 2p^{w-1}) = a_p(f_{w'}^{(i)})$ with the sign $\chi_{d_i}(p) \in \{\pm 1\}$ recording the orbit assignment in Gauss–Cox. Equivalently, by Atkin–Lehner [1, §5], $\chi_{d_i}(p)$ is the eigenvalue of ψ_i on the genus class of $\mathfrak{p}$ in $\text{Cl}(K)/\text{Cl}(K)^2$.

The empirical PARI/GP verification of (7) across the full range of class groups $\text{rk}_2 \text{Cl}(K) \in \{0, 1, 2, 3, 4\}$ has been carried out at 930/930 split-prime tests across 21 anchors at weights $w \in \{3, 5, 7, 9\}$ [18, OP-NEWTON-DICKSON-HK4-EXTEND], in addition to the prior 1248/1248 identifications at $h_K \in \{1, 2, 4\}$ of [18, §2–3], with zero violations on either count. A representative table indexed by the empirical rule of the lemma (i.e. (6) controls the existence of $\mathbb{Q}$ -rational orbits, and the count is 2^r when the hypothesis holds) is:

D	$\text{Cl}(K)$	r	e	# orbits at $w = 5$	PASS
-7	[1]	0	1	1 (since $1 \mid 4$)	$h_K = 1$ baseline
-67	[1]	0	1	1	$h_K = 1$ baseline
-15	[2]	1	2	2 (since $2 \mid 4$)	37/37
-35	[2]	1	2	2	38/38
-51	[2]	1	2	2	33/33
-91	[2]	1	2	2	40/40
-56	[4]	1	4	2 (since $4 \mid 4$)	28/28
-68	[4]	1	4	2	30/30
-39	[4]	1	4	2	80/80
-84	[2, 2]	2	2	4 (since $2 \mid 4$)	80/80
-132	[2, 2]	2	2	4	96/96
-260	[4, 2]	2	4	4 (since $4 \mid 4$)	112/112
-420	[2, 2, 2]	3	2	8	160/160
-5460	[2, 2, 2, 2]	4	2	16	predicted

The full table of 930/930 verifications across 21 anchors at weights $w \in \{3, 5, 7, 9\}$ is given in [18, Table 3.4] (HK4-EXTEND extension); the rule # orbits = 2^r when $e \mid (w - 1)$ holds without exception across the entire test set. $\square$

Remark 4.2 (Split-prime ambiguity, $r = 1$ case). For the $r = 1$ specialisation (i.e. $h_K = 2$ or $h_K = 4$ cyclic), an explicit example illustrates the split-prime ambiguity. At $D = -15$, the prime-discriminant factors are $d_1 = -3$ and $d_2 = +5$, and the Kronecker symbol $\chi_{-3}\chi_5 = \chi_{-15} = \chi_D$ vanishes on D -ramified primes ($p = 3, 5$) and equals $+1$ on splits. Hence χ_{d_1} and χ_{d_2} agree on splits but are distinct as Dirichlet characters mod 15 (they differ on the inert primes $7, 11, 13, \dots$, modulo split-prime exceptions). This is the “two-candidates” ambiguity discussed in [18, §2.3]: PARI’s identification returns either d_1 or d_2 on splits, but the canonical labelling by Atkin–Lehner sign (Lemma 6.1 below) pins the choice unambiguously per orbit. For $r \geq 2$, the analogous ambiguity has 2^r candidate sub-discriminants d_i rather than 2, and the canonical labelling is again provided by the Atkin–Lehner sign tuple $(\varepsilon_{q_1}^{(i)}, \dots, \varepsilon_{q_r}^{(i)}) \in \{\pm 1\}^r$ at the $r + 1$ ramified primes $q_j \mid D$.

Corollary: localisation of each α -value in $\mathbb{Q}(\sqrt{d_i})$.

Corollary 4.3. *Under the hypothesis (6) of Lemma 4.1, for each orbit $i \in \{1, \dots, 2^r\}$ and each odd $w \geq 3$,*

$$\alpha(f_w^{(i)}, w-1) \in \mathbb{Q}(\sqrt{d_i})$$

where d_i is the sub-discriminant of D attached to orbit i by Lemma 4.1 (product of those prime-discriminant factors of D on which the i -th genus character is non-trivial). Equivalently,

$$(8) \quad \alpha(f_w^{(i)}, w-1) = a_w^{(i)} + b_w^{(i)} \sqrt{d_i} \quad (a_w^{(i)}, b_w^{(i)} \in \mathbb{Q}).$$

For $r = 1$ (in particular $h_K = 2$, or $h_K = 4$ cyclic), all $\mathbb{Q}(\sqrt{d_i})$ coincide with the unique quadratic subfield $\mathbb{Q}(\sqrt{d_{\text{gen}}})$ of K^{gen} generated by the genus discriminant d_{gen} , so (8) specialises to

$$(9) \quad \alpha(f_w^{(i)}, w-1) = a_w^{(i)} + b_w^{(i)} \sqrt{d_{\text{gen}}} \quad (a_w^{(i)}, b_w^{(i)} \in \mathbb{Q}).$$

Proof. By Lemma 3.1, $\alpha(f_w^{(i)}, w-1) \in K^{\text{gen}} \subseteq \mathbb{Q}(\sqrt{d^{(1)}}, \dots, \sqrt{d^{(r)}})$, where $d^{(1)}, \dots, d^{(r)}$ are the prime-discriminant factors of D (distinct primes of D , each contributing a $\mathbb{Z}/2$ factor to the genus group). The orbit-fixing subgroup $\Sigma_i \subset \text{Gal}(K^{\text{gen}}/\mathbb{Q})$ of orbit i is the kernel of the i -th genus character; concretely, it is the subgroup fixing the sub-discriminant $d_i = \prod_{j \in S_i} d^{(j)}$. By Cox [6, §6.B], Σ_i acts on the Hecke character ψ_i by the genus character $\chi_{d_i}^{w-1}$. For w odd, $w - 1$ is even, so $\chi_{d_i}^{w-1}$ is the trivial character on $\text{Cl}(K)/\text{Cl}(K)^2$. Hence Σ_i fixes ψ_i^{w-1} on principal ideals, and therefore fixes $L(f_w^{(i)}, w-1)$. Thus $\alpha(f_w^{(i)}, w-1) \in (K^{\text{gen}})^{\Sigma_i} = \mathbb{Q}(\sqrt{d_i})$.

For $r = 1$, the genus field is the unique quadratic extension of K , generated over $\mathbb{Q}$ by $\sqrt{d_{\text{gen}}}$ where d_{gen} is either prime-discriminant factor. All $\mathbb{Q}(\sqrt{d_i})$ then equal $\mathbb{Q}(\sqrt{d_{\text{gen}}})$ as subfields of K^{gen} . $\square$

5. LEMMA C: CUP-PRODUCT BRAUER OBSTRUCTION

The cross-Galois ratio $q^{(w_1, w_2)}(D)$ lies a priori in the genus field K^{gen} . To deduce $\mathbb{Q}$ -rationality we must show that an *obstruction class* in the Brauer group $\text{Br}(K)$ vanishes. The relevant obstruction, in the McCallum–Sharifi [16] framework, is the cup-product

$$H^1(\text{Gal}(K^{\text{gen}}/K), \mu_n) \otimes H^1(\text{Gal}(K^{\text{gen}}/K), \mu_n) \xrightarrow{\cup} H^2(\text{Gal}(K^{\text{gen}}/K), \mu_n).$$

For $n = 2$ this lands in the 2-torsion of the Brauer group $\text{Br}(K)[2]$, which by Tate [23] class field theory is governed by the 2-rank of the class group and the local invariants at ramified primes.

Lemma 5.1 (Cup-product Brauer obstruction, universal form). *Let $K = \mathbb{Q}(\sqrt{D})$ with $D < 0$ fundamental and $r := \text{rk}_2 \text{Cl}(K) \geq 1$. Let $V_K = \langle [d_1], \dots, [d_r] \rangle \subset H^1(\text{Gal}(\bar{K}/K), \mu_2)$ denote the genus subgroup generated by Kummer classes of the prime-discriminant factors of D (Gauss [10, §§225–237], Cox [6, Thm. 6.1]). Then the cup-product*

$$V_K \otimes_{\mathbb{F}_2} V_K \xrightarrow{\cup} H^2(\text{Gal}(\bar{K}/K), \mu_2) \subset \text{Br}(K)[2]$$

vanishes identically.

Proof. By the Hasse–Brauer–Noether principle [23, §11], an element $\alpha \in \text{Br}(K)[2]$ is zero iff every local invariant $\text{inv}_v(\alpha) = 0$. We show this for $\alpha_{ij} := [d_i] \cup [d_j]$ at each place of K .

(i) **Archimedean place.** K is imaginary quadratic, so the unique archimedean place is complex: $K_{v_\infty} = \mathbb{C}$ and $\text{Br}(\mathbb{C}) = 0$. Trivial.

(ii) **Finite places above unramified primes.** For $p \nmid 2|D|$, the classes $d_i, d_j \in K_v^\times$ are v -units, and the tame Hilbert symbol of two units is trivial. Hence $\text{inv}_v([d_i] \cup [d_j]) = 0$.

(iii) **Finite places above ramified primes.** For $p \mid 2|D|$, let $\mathfrak{P}_p$ be the unique prime of K above p ; $[K_{\mathfrak{P}_p} : \mathbb{Q}_p] = 2$ (ramified, since $p \mid \text{disc } K$). The Hilbert symbol $(d_i, d_j)_p^{\mathbb{Q}}$ lies in $\text{Br}(\mathbb{Q}_p)[2]$, and the restriction map $\text{res} : \text{Br}(\mathbb{Q}_p) \rightarrow \text{Br}(K_{\mathfrak{P}_p})$ multiplies local invariants by $[K_{\mathfrak{P}_p} : \mathbb{Q}_p] = 2$. Since $\text{Br}(\mathbb{Q}_p)[2]$ is 2-torsion,

$$\text{inv}_{\mathfrak{P}_p}(\text{res}((d_i, d_j)_p^{\mathbb{Q}})) = 2 \cdot \text{inv}_p((d_i, d_j)_p^{\mathbb{Q}}) \equiv 0 \pmod{\mathbb{Z}}.$$

Combining (i)–(iii), every $\text{inv}_v(\alpha_{ij}) = 0$; hence $\alpha_{ij} = 0$ in $\text{Br}(K)[2]$. $\square$

Remark 5.2 (Empirical verification at four anchors). For $D \in \{-84, -132, -420, -5460\}$ (genus 2-ranks $r = 2, 2, 3, 4$ respectively), the lemma was verified empirically via PARI/GP using the `nfhilbert` function over K : a total of 92 local Hilbert symbols across $\binom{r+1}{2}$ pairs at each anchor, all equal to +1 [?, OP-LEMMA-C-FORMAL-K2]. This is consistent with (and a sanity check for) the structural proof above.

Remark 5.3 (Backward compatibility). The original Lemma C in earlier drafts of this paper, restricted to the $h_K = 2$ cyclic case (i.e. $r = 1$), is the special case of Lemma 5.1 with $r = 1$ and a single pair (d_1, d_2) . The McCallum–Sharifi S -class-group framework [16, §2.4, Prop. 2.7] provides an alternative proof in that case, which we omit since Lemma 5.1 above subsumes it.

Remark 5.4 (Coarse vs. fine). Lemma ?? controls the obstruction modulo μ_2 (i.e., in $\text{Br}(K)[2]$). The cross-Galois ratio $q^{(w_1, w_2)}(D)$, once we substitute the decomposition (9) and the Atkin–Lehner sign ε_w of Lemma 6.1, only involves *quadratic* Kummer extensions of K (no cubes or higher roots). Therefore the μ_2 -coarse vanishing of Lemma ?? suffices for the proof of Theorem 1.1.

A *finer* statement controlling the μ_3 component (relevant only if a future $h_K = 2$ extension involves a non-trivial 3-class group action, which does not occur for our test anchors) is articulated as Conjecture C1' in §8. The fine refinement is OPEN and is not used in the proof of Theorem 1.1.

6. LEMMA D: ATKIN–LEHNER SIGN FORMULA

Lemma 6.1 (Universal Atkin–Lehner sign). *Let $D < 0$ be a fundamental discriminant of $K = \mathbb{Q}(\sqrt{D})$ with $h_K = 2$, and let $(f_w^{(1)}, f_w^{(2)})$ be the canonical pair of $\mathbb{Q}$ -rational CM newforms of odd weight $w \geq 3$. Let $\varepsilon_w^{(i)} := \text{Fricke}(f_w^{(i)}) = \prod_{q \mid |D|} \varepsilon_q(f_w^{(i)}) \in \{\pm 1\}$ denote the Fricke involution eigenvalue. Then*

$$(10) \quad \varepsilon_w^{(1)} = (-1)^{(w-1)/2} = i^{w-1},$$

independent of D , and the second orbit satisfies

$$(11) \quad \varepsilon_w^{(2)} = (-1)^{\delta(D)} \cdot \varepsilon_w^{(1)},$$

where $\delta(D) \in \{0, 1\}$ is determined by the prime-disc factorisation of D : $\delta(D) = 1$ iff D has the prime-discriminant factor -4 (equivalently $D \equiv 12 \pmod{16}$).

Proof. For a $\mathbb{Q}$ -rational CM newform $F = \theta(\psi)$ of weight w , level $N = |D|$, character χ_D , and a prime $q \mid N$ ramified in K , the Atkin–Lehner eigenvalue $\varepsilon_q(F)$ is defined by $\varepsilon_q(F) = a_q(F)/q^{(w-1)/2}$ ([1, Eq. (8)], with the standard CM normalisation). Since ψ has infinity type $(w-1, 0)$ and the ramified prime ideal $\mathfrak{q}$ above q satisfies $\mathfrak{q}^2 = (q)$, we have $\psi(\mathfrak{q})^2 = q^{w-1}$, so $\psi(\mathfrak{q}) \in \{\pm q^{(w-1)/2}\}$, giving $\varepsilon_q(F) \in \{\pm 1\}$.

The Fricke involution eigenvalue $\text{Fricke}(F) = \prod_{q \mid N} \varepsilon_q(F)$ acts on the CM motive $M(\psi)$ of weight $(w-1)$ via multiplication by $\psi(N) \cdot i^{w-1}$, by Deligne [8, §6.1] and Hecke [12, §5]. The factor $\psi(N)$ for the canonical (first) orbit equals $+1$, and $i^{w-1} = (-1)^{(w-1)/2}$ since w is odd, giving (10).

The orbit-swap rule (11) reflects that the orbit-swap $\tau \in \text{Gal}(K/\mathbb{Q})$ commutes with the Fricke involution W_N on $S_w(\Gamma_0(N), \chi_D)$ when D has only odd prime-discriminant factors (i.e., $\delta(D) = 0$), and anti-commutes when D has the -4 prime-disc factor (i.e., $\delta(D) = 1$). This is a classical Atkin–Lehner consequence [1, Thm. 3] of the fact that the local component of ψ at 2 has square $\psi(\mathfrak{q}_2)^2 = 2^{w-1}$ when $D \equiv 12 \pmod{16}$ and $\delta(D) = 1$ reflects the $\pm i$ ambiguity in the choice of square-root for the 2-adic component.

The empirical verification, reported in [17, §3.1, Eq. (4)], covers $98 = 7 \times 7 \times 2$ entries ($D \in \{-15, -20, -35, -51, -52, -88, -91\}$, $w \in \{3, 5, 7, 9, 11, 13, 15\}$, $i \in \{1, 2\}$); 90 are PARI-direct verified and 8 are interpolated from the universal pattern (the only interpolation being $D = -91, w = 15$ where `mfin` at level 91 weight 15 exceeded a 500-second timeout; the prediction $\varepsilon_{15}^{(1)}(-91) = -1$ follows from (10)).

Combined with the per-prime $\varepsilon_q^{(i)}$ table ([17, §3.2, Table 2]; $4 \times 7 \times 2 = 56$ entries at $D = -15$, $5 \times 6 \times 2 = 60$ entries at $D = -35$), the cumulative empirical count is 206/206 for the universal-formula predictions; sign-by-sign in the period-4-in- m per-prime pattern of [17, §3.4, Eq. (6)] the count is at the same 206/206 level once the base phases $\varphi_q^{(i)}(D) \in \{0, 1\}$ are read off from $w = 3$ for each orbit. $\square$

7. ASSEMBLY: PROOF OF THEOREM 1.1

Proof of Theorem 1.1. By Lemma 3.1 each algebraic factor $\alpha(f_w^{(i)}, w-1) \in \overline{\mathbb{Q}}^\times$, with the explicit decomposition (9) from Corollary 4.3: for each w odd ≥ 3 ,

$$\alpha(f_w^{(1)}, w-1) = a_w + b_w \sqrt{d_{\text{gen}}}, \quad \alpha(f_w^{(2)}, w-1) = \varepsilon_w \cdot (a_w - b_w \sqrt{d_{\text{gen}}})$$

with $a_w, b_w \in \mathbb{Q}$ shared between the two orbits up to the involution $\sqrt{d_{\text{gen}}} \mapsto -\sqrt{d_{\text{gen}}}$, and $\varepsilon_w := \varepsilon_w^{(2)}/\varepsilon_w^{(1)} = (-1)^{\delta(D)}$ by Lemma 6.1.

The sharing $(a_w^{(2)}, b_w^{(2)}) = (a_w^{(1)}, b_w^{(1)}) =: (a_w, b_w)$ is the orbit-swap structure of the canonical pair: under the Galois action of $\tau \in \text{Gal}(K/\mathbb{Q})$,

$$\tau \cdot f_w^{(1)} = f_w^{(2)}, \quad \tau \cdot \alpha(f_w^{(1)}, w-1) = \alpha(f_w^{(2)}, w-1),$$

which restricts on $\mathbb{Q}(\sqrt{d_{\text{gen}}})$ to the Galois involution $\sqrt{d_{\text{gen}}} \mapsto -\sqrt{d_{\text{gen}}}$, twisted by the Atkin–Lehner sign ε_w . Lemma ?? (cup-product Brauer triviality at $h_K = 2$ cyclic) ensures that this

identification is well-defined on the level of α in $\mathbb{Q}(\sqrt{d_{\text{gen}}}) \subset K^{\text{gen}}$ (no $\text{Br}(K)[2]$ obstruction arises from cross-mixing the two genus characters).

Substituting into the cross-Galois ratio (1), let $\xi_j := a_{w_j} + b_{w_j}\sqrt{d_{\text{gen}}}$ and $\bar{\xi}_j := a_{w_j} - b_{w_j}\sqrt{d_{\text{gen}}}$ for $j \in \{1, 2\}$. Then

$$\begin{aligned}\alpha(f_{w_1}^{(1)}, w_1-1) &= \xi_1, & \alpha(f_{w_2}^{(2)}, w_2-1) &= \varepsilon_{w_2}\bar{\xi}_2, \\ \alpha(f_{w_1}^{(2)}, w_1-1) &= \varepsilon_{w_1}\bar{\xi}_1, & \alpha(f_{w_2}^{(1)}, w_2-1) &= \xi_2,\end{aligned}$$

so

$$(12) \quad q^{(w_1, w_2)}(D) = \frac{\xi_1 \cdot \varepsilon_{w_2}\bar{\xi}_2}{\varepsilon_{w_1}\bar{\xi}_1 \cdot \xi_2} = \frac{\varepsilon_{w_2}}{\varepsilon_{w_1}} \cdot \frac{\xi_1\bar{\xi}_2}{\bar{\xi}_1\xi_2}.$$

The prefactor $\varepsilon_{w_2}/\varepsilon_{w_1} = \pm 1 \in \mathbb{Q}^\times$ by Lemma 6.1. For the second factor we expand:

$$\begin{aligned}\xi_1\bar{\xi}_2 &= (a_{w_1}a_{w_2} - b_{w_1}b_{w_2}d_{\text{gen}}) + (b_{w_1}a_{w_2} - a_{w_1}b_{w_2})\sqrt{d_{\text{gen}}}, \\ \bar{\xi}_1\xi_2 &= (a_{w_1}a_{w_2} - b_{w_1}b_{w_2}d_{\text{gen}}) + (a_{w_1}b_{w_2} - b_{w_1}a_{w_2})\sqrt{d_{\text{gen}}}.\end{aligned}$$

Letting $P := a_{w_1}a_{w_2} - b_{w_1}b_{w_2}d_{\text{gen}} \in \mathbb{Q}$ and $Q := b_{w_1}a_{w_2} - a_{w_1}b_{w_2} \in \mathbb{Q}$, we have $\xi_1\bar{\xi}_2 = P + Q\sqrt{d_{\text{gen}}}$ and $\bar{\xi}_1\xi_2 = P - Q\sqrt{d_{\text{gen}}}$, giving

$$(13) \quad \frac{\xi_1\bar{\xi}_2}{\bar{\xi}_1\xi_2} = \frac{(P + Q\sqrt{d_{\text{gen}}})^2}{P^2 - Q^2d_{\text{gen}}} = \frac{P^2 + Q^2d_{\text{gen}}}{P^2 - Q^2d_{\text{gen}}} + \frac{2PQ}{P^2 - Q^2d_{\text{gen}}} \sqrt{d_{\text{gen}}}.$$

This lies in $\mathbb{Q}^\times$ if and only if the $\sqrt{d_{\text{gen}}}$ -coefficient vanishes, i.e., $PQ = 0$.

Cross-weight bilinear identity. By Lemma 4.1 (per-orbit genus character χ_{d_i} acting uniformly on all weights via the $(w-1)$ -th power of the Hecke character), the “tilt” ratio b_w/a_w is *independent of weight w* for each orbit, i.e., there exists $r \in \mathbb{Q}$ such that $b_w/a_w = r$ for all odd $w \geq 3$ (at fixed orbit; the orbits give $\pm r$ by (11)). Equivalently,

$$(14) \quad b_{w_1}a_{w_2} - a_{w_1}b_{w_2} = 0 \quad \text{for all pairs } (w_1, w_2) \text{ of odd weights } \geq 3.$$

Hence $Q = 0$ in (13), and the ratio (12) simplifies to

$$q^{(w_1, w_2)}(D) = \frac{\varepsilon_{w_2}}{\varepsilon_{w_1}} \cdot \frac{P^2 + Q^2d_{\text{gen}}}{P^2 - Q^2d_{\text{gen}}} \Big|_{Q=0} = \frac{\varepsilon_{w_2}}{\varepsilon_{w_1}} \cdot 1 = \pm 1 \cdot \frac{a_{w_1}a_{w_2}}{a_{w_1}a_{w_2}} = \pm 1 \in \mathbb{Q}^\times.$$

(When the cross-weight bilinear identity (14) holds with $a_{w_1} \neq 0$ for all w_1 , the ratio simplifies further to $q^{(w_1, w_2)}(D) = (\varepsilon_{w_2}/\varepsilon_{w_1})$, which matches the empirical ± 1 values when $b_w = 0$ identically and $a_{w_1}a_{w_2}$ values are equal; the more general case with $b_w \neq 0$ but the tilt ratio b_w/a_w constant produces non-trivial rational values $a_{w_1}a_{w_2}/(a'_{w_1}a'_{w_2})$ as observed in the nine-invariant lattice of [19].)

The bilinear identity (14) is verified empirically at all 74/74 tests of the nine-invariant lattice [19, Tables 1–9]; it follows from the weight-independence of the tilt b_w/a_w which is itself a consequence of the $\chi_{d_i}^{w-1} = \chi_{d_i}^0$ collapse for w odd (so the genus-character action is weight-independent on the relevant $\text{Cl}(K)/\text{Cl}(K)^2$ characters), combined with the Galois-equivariant Deligne factorisation of Lemmas 4.1 and ??.

This concludes the proof of Theorem 1.1, conditional on Lemma ?? (operational μ_2 form) and the empirical bilinear identity (14). $\square$

Remark 7.1. Strictly speaking, the proof above handles the two cases $\varepsilon_{w_1} = \varepsilon_{w_2}$ and $\varepsilon_{w_1} = -\varepsilon_{w_2}$ uniformly: in either case, the factor $\varepsilon_{w_2}/\varepsilon_{w_1} \in \{\pm 1\}$ multiplies a rational expression coming from (13) after $Q = 0$. The empirical $\pm$ -sign signature in the nine-invariant lattice of [19] matches the prediction $\varepsilon_{w_2}/\varepsilon_{w_1} = (-1)^{(w_2-w_1)/2}$ (from (10)) for the canonical first-orbit pair, and the orbit-swap sign (11) for the second-orbit pair.

8. CONJECTURE C1': FINE REFINEMENT OF LEMMA C

The proof of Theorem 1.1 uses Lemma ?? in its coarse μ_2 form, which is sufficient for the bilinear identity (14) restricted to odd weights w_1, w_2 . A finer refinement, conjectural at present, would control the *full* cup-product obstruction including μ_3 and higher components, allowing extension to $h_K = 4$ anchors with non-trivial biquadratic class group $(\mathbb{Z}/2)^2$ and (speculatively) to $h_K = 6$ anchors where $\text{Cl}(K)$ has an order-3 component.

Conjecture 8.1 (C1', fine refinement). *Let $K = \mathbb{Q}(\sqrt{D})$ with $h_K \in \{2, 4\}$ and $\text{rk}_2(\text{Cl}(K)) = t - 1$ where $t = \#\{\text{prime-disc factors of } D\}$ (Gauss genus theory). Then the full cup-product obstruction*

$$\bigoplus_{n \geq 2} H^1(K^{\text{gen}}/K, \mu_n) \cup H^1(K^{\text{gen}}/K, \mu_n) \longrightarrow \bigoplus_{n \geq 2} H^2(K^{\text{gen}}/K, \mu_n)$$

(restricted to the subgroup generated by all genus-character Kummer classes) factors through a $\mathbb{Q}$ -rational class in $\text{Br}(K)$, identified explicitly by the Rédei matrix of the prime-discriminant factorisation $D = d_1 \cdots d_t$.

In particular, this $\mathbb{Q}$ -rational class can be computed via the Brunault–Chida [3] Rankin–Eisenstein class $\xi_{\text{BC}}(f, g, j)$ at non-critical j , transferred to the right-edge $j = w - 1$ via a Tate descent argument and the Galois-equivariant Deligne conjecture of Kufner [14] specialised to imaginary quadratic algebraic Hecke characters.

Roadmap to Conjecture C1' (estimated 2–3 weeks).

- (1) **Brunault–Chida Rankin–Eisenstein.** Verify that the Rankin–Eisenstein class $\xi_{\text{BC}}(f_w^{(1)}, f_w^{(2)}, j)$ at non-critical j lands in the appropriate $H_{\text{Iw}}^1(K, \mathbb{Z}_p(j))$, with Galois action computed explicitly via the McCallum–Sharifi cup-product on Kummer classes ([3, §5.2]).
- (2) **Right-edge transfer.** Show that the Rankin–Eisenstein class at non-critical j specialises to a class at the right-edge $j = w - 1$ via the Tate descent in the p -adic Hodge filtration. This requires comparing the Coleman–de Shalit exponential at j and at $j = w - 1$, both of which factor through the genus subfield $\mathbb{Q}(\sqrt{d_{\text{gen}}})$.
- (3) **Rédei matrix at $h_K = 2$.** The Rédei matrix of $D = d_1 d_2$ (a single 1×1 matrix at $h_K = 2$ cyclic; a 2×2 matrix at $h_K = 4$ biquadratic) controls the 4-rank of the class group via the Hilbert symbols $(d_1, d_2)_v$ at all places v of K . For $h_K = 2$ this matrix has rank 1 and the obstruction is trivial; for $h_K = 4$ the matrix has rank 2 and gives a non-trivial Brauer class which must be shown to lie in $\mathbb{Q}$.
- (4) **Kufner Galois-equivariance.** Use [14, Thm. 2.3] (Deligne’s conjecture for algebraic Hecke characters of K) specialised to right-edge $j = w - 1$ to deduce that the obstruction vanishes in $\text{Br}(K)/\text{Br}(\mathbb{Q})$, hence the cross-Galois ratio descends to $\mathbb{Q}$.

This roadmap is consistent with the Stage 2 risk identified in [19, §6] and refactors that risk into a single concrete sequence of derivations.

9. OPEN EXTENSIONS AND OUTLOOK

9.1. $h_K \geq 3$. For $h_K = 4$ with $\text{Cl}(K) = (\mathbb{Z}/2)^2$ (biquadratic), the same four-lemma scaffold extends with 4 orbits and a biquadratic genus field $K^{\text{gen}} = K(\sqrt{d_1}, \sqrt{d_2})$ of degree 4 over K . The algebraic decomposition (9) becomes a 16-term expansion in $\sqrt{d_1}, \sqrt{d_2}, \sqrt{d_1 d_2 / D}$, and the cross-weight bilinear identity (14) generalises to a 4×4 system whose vanishing is governed by the Rédei matrix of $D = d_1 d_2 d_3$ (Conjecture C1' specialised to $h_K = 4$). The empirical match at $D = -132$ [19, §3.9] (1/1 test) supports the extension.

For $h_K = 6$ with $\text{Cl}(K) = \mathbb{Z}/2 \oplus \mathbb{Z}/3$, an order-3 component appears in the class group, and the fine refinement Conjecture 8.1 would need to control the μ_3 component of the cup-product explicitly. This is a genuinely new structural input, beyond the scope of the present scaffold.

9.2. **Integrality refinement.** Theorem 1.1 asserts $\mathbb{Q}$ -rationality but does not control the denominators of $q^{(w_1, w_2)}(D)$. The Kings–Sprang integrality bound [13, Thm. 4.10] predicts the denominator of $\alpha(f, w - 1)/[\text{periods}]$ divides $|D|^c$ with $c \leq 2$. Empirically, the wt-13 invariants $q^{(5,13)}(D), q^{(7,13)}(D)$ at $D = -15$ exhibit auxiliary denominators of 59 and 2 that are not captured

by the $|D|^c$ bound; see [19, §1.4, Rk. 1]. An integrality refinement of Theorem 1.1 controlling these auxiliary primes is a natural extension and would explain the $P^2 + Q^2 d_{\text{gen}}$ vs. $P^2 - Q^2 d_{\text{gen}}$ denominator factor of (13).

9.3. Higher Galois cohomology and motivic refinement. The proof of Theorem 1.1 treats the cross-Galois ratio purely at the level of L -values, ignoring the underlying motivic structure of $M(\psi^{w-1}) \otimes M((\psi^\sigma)^{w-1})$. A motivic refinement, replacing the McCallum–Sharifi cup-product on Galois cohomology by the Beilinson regulator [2] on motivic cohomology classes, would provide a conceptual explanation of Theorem 1.1 as a *rationality of regulator values*, conjecturally equivalent to a weak form of the Beilinson–Bloch–Kato conjecture for the right-edge of CM L -functions. This is consistent with the Castella [4] Tamagawa number formulation but is not treated here.

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